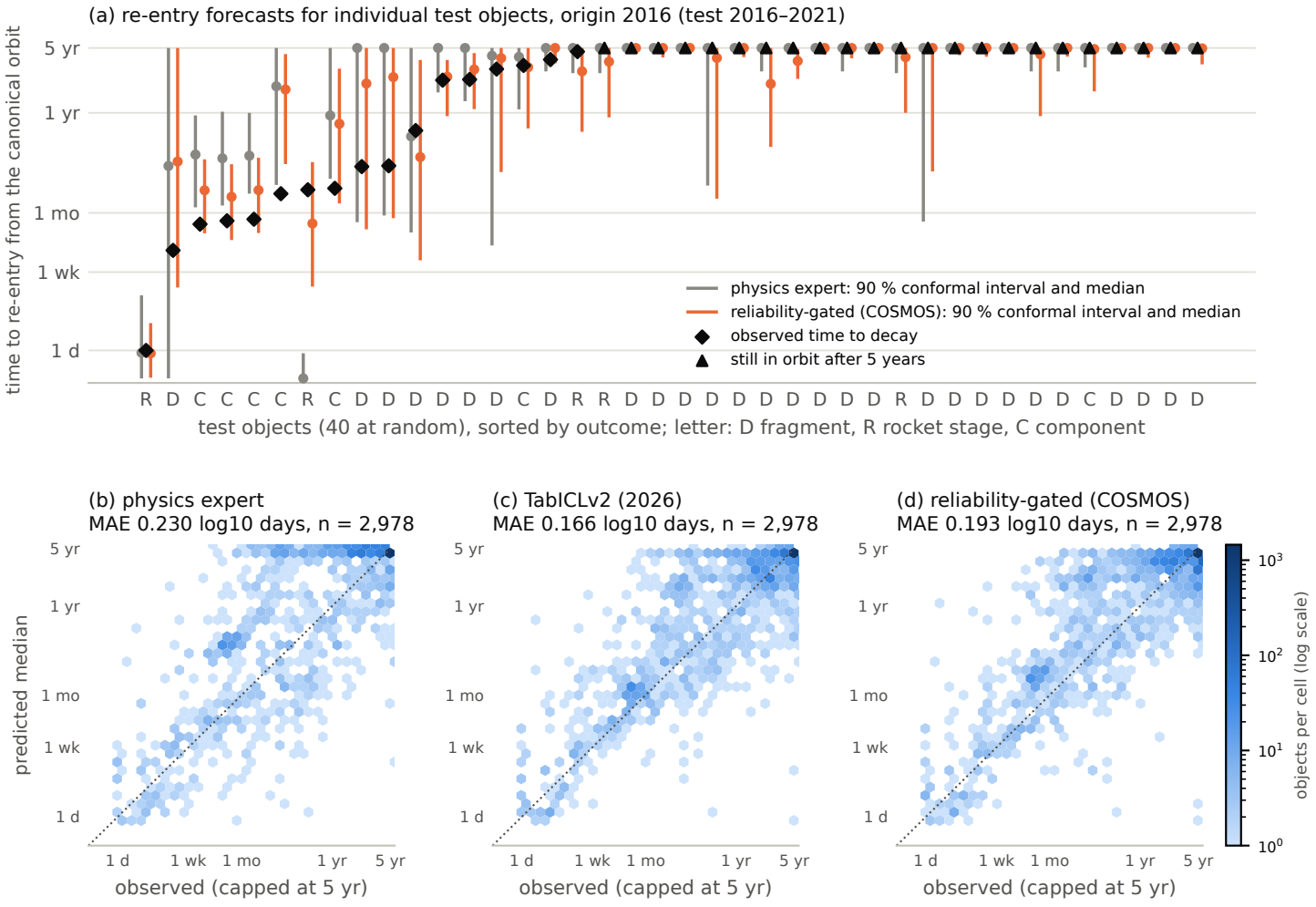
(a) re-entry forecasts for individual test objects, origin 2016 (test 2016–2021)
time to re-entry from the canonical orbit
5 yr
1 yr
1 mo
1 wk
1 d
physics expert: 90 % conformal interval and median
reliability-gated (COSMOS): 90 % conformal interval and median
observed time to decay
still in orbit after 5 years
R D C C C C R C D D D D D D C D R R D D D D D D D D D R D D D D D D C D D D D
test objects (40 at random), sorted by outcome; letter: D fragment, R rocket stage, C component
(b) physics expert
MAE 0.230 log10 days, n = 2,978
(c) TabICLv2 (2026)
MAE 0.166 log10 days, n = 2,978
(d) reliability-gated (COSMOS)
MAE 0.193 log10 days, n = 2,978
predicted median
observed (capped at 5 yr)
1 d
1 wk
1 mo
1 yr
5 yr
objects per cell (log scale)
10^3
10^2
10^1
10^0